

Capitalism and Technical Change

In section 5.2, we had studied the issue of technical change, and we now revisit the issue to deal with it in greater detail. In particular, two questions related to technical change will be addressed in this chapter. We will first make precise the definition of technical change, relate it to capitalist decision-making and then ask the following question: Does technical change adopted by capitalist firms coincide with what is best from a social perspective? The second question we will investigate relates to the relationship between technical change and the rate of profit. When does technical change in capitalist economies lead to a fall in the rate of profit?

The second question derives its importance from a long and distinguished Marxist literature on tendencies of economic crisis in capitalist economies.¹ From a Marxist perspective, *an economic crisis in a capitalist economy is a deep and prolonged interruption of the economy-wide circuit of capital*. The proximate cause of an economic crisis is a decline in the average rate of profit. The rate of profit can fall in two different, and mutually exclusive, ways, which suggest a typology of economic crises in capitalism. The first way in which the average rate of profit can fall is when the economy is marked by a chronic insufficiency of aggregate demand, so that commodities are sold at prices that are below their value (or price of production). Hence, in this case, the sale of the commodity does not realize the full surplus value (or the average rate of profit), and the realized rate of profit falls below the ‘normal’ rate of profit (which prevailed previously). We can refer to this as a ‘crisis of excess surplus

¹For a discussion of economics crisis and different Marxist explanations of the important crisis tendencies, see Basu (2019).

value' (because more surplus value was produced than could be realized through sale). The second way in which the average rate of profit can fall is when, despite the commodity selling at its full value (or price of production), the realized rate of profit declines. Thus, in this case, the problem is not one of realization of the surplus value embedded in commodities, but rather points to the production of insufficient surplus value itself. We can refer to this as a 'crisis of deficient surplus value' (because the system produces less surplus value than is necessary to ensure a 'normal' rate of profit).²

Two mechanisms that can account for the crisis of deficient surplus value are, first, the profit-squeeze mechanism (discussed in Chapter 25, Volume I of *Capital*) and, second, the rising OCC mechanism (discussed in Chapter 13, Volume III of *Capital*). The second mechanism is associated with Marx's law of the tendential fall in the rate of profit (LTFRP) and claims to establish the proposition that the average rate of profit has a tendency to fall with capitalist development due to rising mechanization of production, the latter being a key feature of capitalist technical change. In a famous contribution, Okishio (1961) was seen as overturning Marx's result and establishing that technical change would, in fact, lead to a rise in the average rate of profit. In this chapter, we will study the relationship between Marx's original claim and Okishio's result – both of which relate to the relationship between technical change and the rate of profit in capitalist economies. We will want to understand which position, Marx's or Okishio's, is right? Is Marx's LTFRP overturned by Okishio? In this chapter, we will see that there is less of a contradiction between Marx's and Okishio's results. We will see that both results can be reconciled in a more general framework, which includes technical change and class struggle. The main result in this respect that we want to derive and highlight is that depending on the state of class struggle, technical change can lead to either a fall or a rise in the rate of profit, a result we had briefly discussed in section 5.2 earlier.

²What I have called a normal rate of profit has been called the 'usual' rate of profit by Marx and Sweezy. The important point is that when the actual rate of profit falls below the 'usual' rate of profit in more or less all industries in an economy, the rate of capital accumulation slows down and can initiate a period of economic crisis (Sweezy, 1942, p. 142).

6.1 Technical Change

6.1.1 Basic Set-Up

To develop some of the key Marxist arguments relating to capitalism and technical change, I will use a model of a capitalist economy. This model capitalist economy produces only one commodity. Let us call this commodity, corn, and note that it can be used both for consumption and investment.³ Such a model of the capitalist economy is very much like the model used by David Ricardo or like the one used in modern macroeconomic growth theory. It is decidedly simple, and the simplicity has both advantages and disadvantages. The advantage is its accessibility, and the disadvantage is that some issues, for example, relative price changes, cannot be studied in this model. For the purposes of discussing issues of technical change, the advantage seems to outweigh the disadvantage because many of the important aspects of technical change can be conveyed in this single-commodity model.⁴

In the one-commodity model that we are using, the inputs to production are saved-up corn and labour. Saved-up corn is owned by capitalists and labour is provided by workers – who sell their labour-power to capitalists for a money wage rate, w . Workers use the money wage to purchase and consume b units of corn per hour of labour. Hence, if the price of corn is p , then the real wage rate is related to the nominal wage rate as follows: $b = w/p$.

In this economy, a *technique of production* is a combination of two numbers: the amount of corn and the amount of labour that is needed to produce 1 unit of corn. For instance, a technique of production could be represented by the pair of numbers (n, a) , in which case it would mean that n units of labour and a units of corn are needed to produce 1 unit of corn. We assume that labour is indispensable for production, so that $n > 0$, but the non-labour input can be dispensed with, so that $a \geq 0$. For the technique of production to be technologically meaningful, it must produce a positive net output, that is, $1 - a > 0$, so that $a < 1$. Otherwise, the technique of production would use up more inputs than it produces, which is clearly unviable from a purely technological viewpoint.

³Recall that we have already encountered this model in section 3.1.5.

⁴Many of the arguments developed in this section have been treated in multi-commodity models of production. For a two-commodity treatment, see Laibman (1992); for a n -commodity treatment, see Roemer (1981).

By *technical change*, we will mean the availability of new techniques of production. How new techniques of production become available is an important and interesting question, but we will not pursue it here. We will investigate the consequences of capitalist relations of production on the adoption of the new techniques of production made available by technical change. To get started, we must define certain aspects of technical change: productiveness, progressiveness, viability (in a capitalist sense) and types of technical change.

6.1.2 *Productiveness*

Suppose a technique of production is given by (n, a) , that is, n units of labour and a units of corn are needed to produce 1 unit of corn. We will say that this technique of production is *productive* if the total output of corn is larger than the total input of corn, where the latter is the sum of the *direct* input in production as capital, a units of corn, and the *indirect* input as consumption requirement of workers, nb units of corn. Hence, the condition that makes the technique of production (n, a) productive is given by

$$a + nb < 1 \quad (6.1)$$

The above condition is telling us that a technique of production is productive when it can produce a *net* output, that is, total output less the non-labour inputs, $1 - a$, that is, larger than the consumption requirements of the workers, nb . Since labour input is indispensable for production, that is, $n > 0$, and workers need to consume some positive amount of corn per hour of labour, that is, $b > 0$, a necessary condition for the productiveness of the technique of production is that $a < 1$. That is why we have assumed that $a < 1$, which ensures that the technique of production is not only technologically viable but also productive.

The intuition behind the definition in (6.1) is straightforward. The excess of the net output over the consumption needs of the workers is potentially available for reinvestment to improve technologies of production and ensure technological progress over time. If techniques of production were not productive in this sense, society would either stagnate (when the above relationship holds as an equality) or disintegrate and disappear (when the above inequality is reversed).⁵

⁵To keep the analysis simple, we are abstracting from consumption needs of capitalists, but that can be

6.1.3 *Progressiveness*

To make precise the notion of progressive technical change, let λ and λ' denote the value of, that is, labour embodied in, one unit of corn when the techniques of production are (n, a) and (n', a') , respectively. When the technique of production (n, a) is used, a units of corn is combined with n units of labour to produce 1 unit of corn. Hence, the equation for determining the value of a unit of corn can be written as $\lambda = \lambda a + n$ and, by a similar argument, $\lambda' = \lambda' a' + n'$. We can rewrite the equations to derive expressions for the value of corn as $\lambda = n/(1 - a)$ and $\lambda' = n'/(1 - a')$. Technical change is defined to be *progressive* when $\lambda' < \lambda$, that is, when the value of corn falls. A little algebraic manipulation shows that the following condition ensures that technical change is progressive:

$$a' < 1 - \left(\frac{1 - a}{n} \right) n' \quad (6.2)$$

What is the idea behind this definition? It is a way to make concrete the notion that reducing the overall toil of labour is in the general interest of society. When technical change is progressive it reduces the total, that is, sum of direct and indirect, labour content of commodities. Hence, the same amount of commodities can be produced with a smaller amount of labour or a larger amount of commodities can be produced with the same amount of labour. This opens up the possibility of supporting a higher material standard of living and also potentially increasing the time available for leisure. It is these important and worthy societal goals that are captured by the definition of a progressive technical change. The inequality in (6.2) gives the precise quantitative relationship between the two techniques of production that must hold if we are to have progressive technical change: net output per unit of the labour input must be higher in the new technique of production compared to the old.

6.1.4 *Capitalist Viability*

We have seen in previous chapters, especially in Chapter 3, that capitalism's driving force and primary motivation is rooted in profitability. The need to generate and realize surplus value structures all other aspects of the system,

easily added without changing the qualitative features of the analysis.

including technical change. When capitalist producers decide whether to adopt a new technique of production that has become available, their choice is based on cost and profitability calculations. It is this key idea that is captured by the definition of capitalist viability.

We will call the new technique of production, (n', a') , to be *capitalistically viable* if it reduces the cost of production at the *current* price and real wage rate. If the old technique of production, (n, a) were to be used, and the existing price is p , then the cost of producing one unit of output would be $pnb + pa$. On the other hand, if the new technique were to be used at the same existing price, p , the cost of production would be $pn'b + pa'$. Thus, the new technique of production is capitalistically viable if and only if the following condition holds:

$$bn' + a' < bn + a \quad (6.3)$$

There are at least two conceptual motivations for introducing this definition. To begin with, the definition is a realistic, though simplified, description of actual capitalist behaviour. It seems realistic to assume that capitalist producers will be governed by cost and profitability calculations in choosing new techniques of production. But there is a deeper reason too. It is a fundamental Marxist critique of capitalism that profitability calculations are not always aligned with broader societal goals. In the domain of technical change, new techniques of production that are progressive might not be adopted by capitalist producers because they do not reduce the cost of production at current prices. If this were to happen, we would immediately see a gulf opening up between what is best for capitalism (captured by capitalist viability) and what is best for society (captured by progressiveness). To investigate if and when this might happen is the second motivation for introducing the above definition of capitalist viability (and distinguish it from progressiveness). But before we investigate this issue, let us introduce some terminology about types of technical change.

6.1.5 *Types of Technical Change*

We can define different types of technical change from a purely quantitative perspective regarding the two inputs that make up a technique of production. Comparing the old and new techniques of production, (n, a) and (n', a') , we see that there are four distinct possibilities – depending on whether the capital-

input and labour-input requirement is smaller (or larger) in the old versus the new technique of production: $n' > n$ and $a' > a$; $n' > n$ and $a' < a$; $n' < n$ and $a' > a$; $n' < n$ and $a' < a$.⁶ To identify these four types of technical change, let us introduce some terminology. Technical change is of the type

- capital-using, labour-using (CU-LU), if $a' > a$ and $n' > n$, that is, more capital and more labour is needed to produce one unit of output with the new technique of production;
- capital-saving, labour-using (CS-LU), if $a' < a$ and $n' > n$, that is, less capital and more labour is needed to produce one unit of output with the new technique of production;
- capital-using, labour-saving (CU-LS), if $a' > a$ and $n' < n$, that is, more capital and less labour is needed to produce one unit of output with the new technique of production;
- capital-saving, labour-saving (CS-LS), if $a' < a$ and $n' < n$, that is, less capital and less labour is needed to produce one unit of output with the new technique of production; a special case of the CS-LS type, when the degree of capital-saving and labour-saving is the same, is known as *Hicks neutral* technical change;
- pure labour-saving, also known as *Harrod neutral*, if $a' = a$ and $n' < n$, that is, same amount of capital and less labour is needed to produce one unit of output with the new technique of production;
- pure capital-saving, also known as *Solow neutral*, if $a' < a$ and $n' = n$, that is, less capital and same amount of labour is needed to produce one unit of output with the new technique of production.

Leading capitalist countries have almost always witnessed labour-saving technical change, that is, the labour productivity has risen over time (Foley et al., 2019, Table 2.8). For some periods, the growth of labour productivity has come along with a fall in capital productivity, so that we have had CU-LS technical change; at other periods, both labour and capital productivity have increased, so that we have had CS-LS technical change.⁷ In the history of capitalism, periods of CU-LS and CS-LS types of technical

⁶I have ignored the various cases of equalities in these relationships because they are not interesting from the perspective of technical *change*.

⁷Note that, with reference to a technique of production (n, a) , $1/n$ is the labour productivity, and $1/a$ is the capital productivity.

change have alternated, with the former dominating the latter (Foley and Michl, 1999). Over the last few decades, some countries have also witnessed CS-LU technical change, especially in the context of deindustrialization. In these cases, the shrinking of the industrial sector relative to the other sectors has probably pushed labour into low-productivity and low-paying service sector jobs in the informal sector, so that overall labour productivity has fallen.⁸

From the perspective of capitalists, CS-LS is the most favourable type of technical change because it saves on both inputs. From a theoretical perspective, the CU-LS type of technical change is more interesting because it highlights certain trade-offs that capitalists face with respect to the choice of technique. These trade-offs open up the possibility of the divergence between not only social and capitalist goals but also between goals of individual capitalists and the capitalist class as a whole. Before looking at these important issues, it might be useful to pause and think about how one might define the types of technical change that we have discussed here, for example, CU-LS technical change, in a more general set-up with many commodities.

In a multi-sector circulating capital model of an economy with n sectors (and each sector producing only one commodity), each technique of production will be a vector of size $n + 1$. The first n elements of this vector will be the material inputs, that is, physical quantities, of the n commodities needed to produce one unit of the commodity; the last element will be the magnitude of labour input (assuming that labour is homogeneous). In such a case, there is no unique way to define, for instance, capital-using technical change. This is because there is no unique way to compare the sizes of the two n vectors of required material inputs before and after technical change. Following from the work of Morishima (1973), one approach to defining capital-using technical change has been the following: a new technique is considered to be capital-using if *every* element of the material input vector changes by non-negative amounts. While this captures the intuitive notion of ‘more’ capital being used in the new technique, it is clear that the definition is very restrictive (because *every* material input must rise or remain unchanged). There can be other ways to implement the intuition of ‘more’ capital input, for example, a larger Euclidean norm of the new material input

⁸For an excellent discussion of this and other issues related to technical change in capitalist economies, see Foley et al. (2019, Chapter 8).

vector, or a larger magnitude of some aggregate measure of the n vector of material inputs. One crucial advantage of working in a one-commodity model, as we have done in this section, is that we avoid this ambiguity – in a one-sector model, types of technical change can be defined uniquely and unambiguously.

6.2 Progressive Technical Change and Capitalism

The first question we wish to investigate is the relationship between progressive technical change and capitalist viability. For this analysis, we will use the diagram in Figure 6.1. In this diagram, we measure the labour input requirement, n' , and the capital input requirement, a' , associated with a *new technique of production* on the horizontal and vertical axes, respectively. The original technique of production, (n, a) , is represented by the point O' .

The straight line AB represents the boundary (or limiting case) of progressive technical change given in (6.2). Hence, any point inside the triangle AOB represents progressive technical change, that is, they represent new techniques of production that reduce the labour embodied in a unit of corn compared to the old technique of production O' . On the other hand, the straight line CD represents the capitalist viability condition in (6.3). Hence, any point inside the triangle COD represents new techniques of production that reduce the total cost of producing a unit of corn at the existing price and wage rate, compared to the old technique of production O' . Two important results that highlight the possible technological irrationality of capitalism follow immediately from an inspection of Figure 6.1.

The first result is that there are many instance of progressive technical changes that will *not* be adopted by capitalist producers because they are not capitalistically viable. Starting from the original technique of production O' , the whole set of progressive technical changes that will not be adopted by capitalist producers is represented by the triangle $AO'C$. This triangle represents all new techniques of production that are technologically progressive (because they are inside the triangle AOB) but are not capitalistically viable (because they are outside the triangle COD).

The second result is that capitalist producers might adopt new techniques of production that are *technologically regressive*. Starting from the original

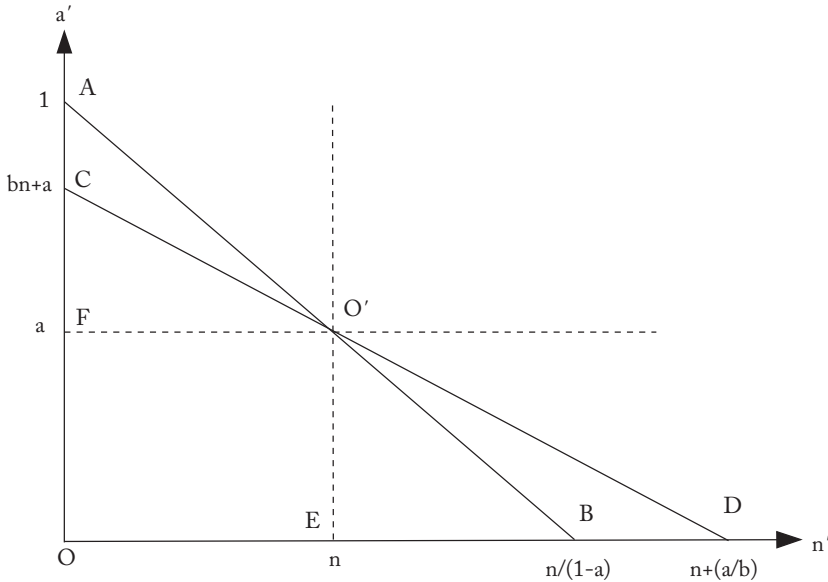


Figure 6.1 The relationship between progressive technical change and capitalistically viable technical change

Source: Author.

technique of production O' , the set of technologically regressive techniques of production that might be adopted by capitalist producers is represented by the triangle $BO'D$. This triangle represents all new techniques of production that are capitalistically viable (because they lie inside the triangle COD) but are technologically regressive (because they are outside the triangle AOB).

Let us probe further. Using the original technique of production O' , we can divide up the set of all the possible new techniques of production into four groups. These groups are represented by the regions to the northeast, southeast, southwest and northwest of O' . Northeast, northwest, southwest and southeast regions represent CU-LU, CU-LS, CS-LS and CS-LU technical change, respectively. From our discussion earlier, we know that the only really interesting case is the CU-LS technical change. Hence, we need to only focus on the region to the northwest of O' .

When we focus on the region to the northwest of O' , we see another interesting result: for CU-LS type of technical change, all capitalistically viable techniques of production are also technologically progressive. To see

this, note that the CU-LS type of technical change that involves capitalistically viable techniques of production (compared to the original one) is represented by the triangle $CO'F$; and CU-LS type of technical change that involve technologically progressive techniques of production (compared to the original one) is represented by the triangle $AO'F$. Since the triangle $CO'F$ is completely contained in the triangle $AO'F$, the result immediately follows that if we restrict ourselves to only CU-LS type of technical change, then capitalistically viable techniques of production are technologically progressive.⁹

6.3 Technical Change and the Rate of Profit

Now we turn to investigating the second question: What is the impact of technical change on the rate of profit? As I have noted in the introduction to this chapter, this question has attracted enormous scholarly attention over the past several decades.¹⁰ Using Marx's analysis of LTFRP in Volume III of *Capital*, many scholars have understood the claim that technical change can lead to a fall in the rate of profit as a central proposition of Marxist political economy. After several decades of work on this issue, we have a fairly clear picture (Okishio, 1961; Roemer, 1981; Foley, 1986b; Laibman, 1992). Technical change cannot, on its own, lead to a fall in the rate of profit in a competitive setting if capitalists are assumed to behave in a reasonable manner, that is, if they choose to adopt new techniques if and only if it reduces the cost of production at existing prices. But the interaction of technical change and class struggle over the distribution of income *can* lead to a fall in the rate of profit. This is the main result I want to highlight in this section.

When the original technique of production, (n, a) , is used, the rate of profit is given by

$$r = \frac{\text{profit}}{\text{total outlay}} = \frac{p - p(nb + a)}{p(nb + a)} = \frac{1 - nb - a}{nb + a}$$

⁹All the results of this section have been proved in a n -commodity model in Roemer (1977).

¹⁰For an introduction to the discussion of technical change and the rate of profit, see section 5.2.

where r is the rate of profit and p is the price of the commodity. A little algebraic manipulation shows that the rate of profit is given by

$$r = \frac{1}{nb + a} - 1 \quad (6.4)$$

The relationship in (6.4) can be called the *real wage rate-profit rate frontier*. It shows that, given the technique of production, (n, a) , there is a negative relationship between the real wage rate, b , and the rate of profit, r . If the real wage rate rises, the rate of profit must fall, and vice versa. Thus, the income of the working class and the income of the capitalist class are negatively related to each other. That is why the relationship in (6.4) highlights the source of distributional conflict between the fundamental social classes.¹¹

Suppose a new technique of production becomes available to an innovator capitalist that can produce 1 unit of corn using a' units of corn and n' units of labour, that is, the new technique of production is (n', a') . The innovator capitalist would only consider the new technique of production for adoption if it is capitalistically viable, that is, it reduces the cost of production at existing prices. Hence, the new technique of production, if it is to be adopted, must satisfy the condition given in (6.3). This will immediately imply that the innovator capitalist can earn a higher rate of profit if she adopts the new technique of production. To see this, note that if the new technique of production is used in production, it will generate a rate of profit, r' , where

$$r' = \frac{\text{profit}}{\text{total outlay}} = \frac{p - p(n'b + a')}{p(n'b + a')} = \frac{1 - n'b - a'}{n'b + a'}$$

Hence, the rate of profit associated with the new technique of production is

$$r' = \frac{1}{n'b + a'} - 1 \quad (6.5)$$

so that, we have

$$r' - r = \frac{1}{n'b + a'} - \frac{1}{nb + a} \quad (6.6)$$

¹¹ A slightly different way of writing the real wage-profit rate frontier gives us what Foley and Michl (1999) have called the growth-distribution schedule. For a discussion of the real wage rate-profit rate frontier in an economy with more than one commodity, see Appendix A in Chapter 7.

Since the viability condition (6.3) must hold, this implies that $r' > r$.

The higher rate of profit provides the incentive to the innovator capitalist to adopt the new technique of production, (n', a') . Over time, the new technique, which generates a higher rate of profit, diffuses through the capitalist economy. Eventually, all capitalist producers adopt the new technique of production. When this happens, a new price of the commodity emerges, p' . At this new price, the average rate of profit in the economy now becomes

$$r'' = \frac{\text{profit}}{\text{total outlay}} = \frac{p' - p'(n'b + a')}{p'(n'b + a')} = \frac{1 - n'b - a'}{n'b + a'} = r' > r$$

Thus, we have a striking result: technical change that is capitalistically viable leads to a rise in the average rate of profit. In simplified form, this is the essence of the celebrated Okishio theorem (Okishio, 1961). In the one-commodity model we have been using, we have $r'' = r'$, that is, the average rate of profit after technical change was adopted by all capitalists is the same as the rate of profit earned by the innovator capitalist when she first introduces the new technique of production. That is why demonstrating the increase in the average rate of profit was easy. In a n -commodity model, like the one used in Okishio (1961), in general, the two profit rates will not be the same, that is, $r'' \neq r'$, and yet it can be demonstrated that $r'' > r$. In fact, the implication runs the other way too: if the average rate of profit rises after technical change, it must be capitalistically viable.¹²

The result described by the Okishio theorem shows that progress of capitalism will lead to a rising rate of profit.¹³ It seems to contradict Marx's equally celebrated claim that the progress of capitalism will lead to a falling rate of profit (Marx, 1993b). In fact, there is less of a contradiction than seems at first sight.

6.4 A Marx-Okishio Threshold

The validity of the results of the Okishio theorem rests on a crucial, and implicit, assumption – that the real wage rate, b , remains unchanged after the new technique of production has been adopted widely in the capitalist

¹²This result is easy to see in the one-commodity model that we are using. It was first proved in a multi-sector model by Dietzenbacher (1989, Theorem 2).

¹³For an elegant proof of this theorem in a n -commodity model, see Roemer (1981, Theorem 4.6).

economy. In a sense, therefore, the result of Okishio's theorem is not surprising: if the real wage rate remains unchanged, then all the benefits of technological progress in terms of productivity growth are taken by capitalists. Naturally, then, the rate of profit rises.

While the assumption of a constant real wage rate might be true in some scenarios (for example, in a labour-surplus economy), it might not hold in others (for example, in a labour-constrained economy). Thus, it is interesting to revisit the Okishio theorem without this restrictive assumption. It seems intuitively clear that whether or not the rate of profit falls after the adoption of a new technique of production would depend on the magnitude of the growth of the real wage rate. After all, if the rate of profit rises with a constant real wage rate, it is equally clear that it will fall if the real wage rate rises by a large magnitude.

The idea that the rate of profit will fall even when technical change is capitalistically viable if the real wage rate rises fast enough is well known in the literature (Okishio, 1961; Roemer, 1977, 1981; Foley, 1986b; Dietzenbacher, 1989; Laibman, 1992). Once we recognize the dependence of Okishio's result on the fixity of the real wage rate, its critical thrust is significantly weakened. Since we are working with a one-commodity model, we can go one step further and derive what can be called a Marx-Okishio threshold (Basu, 2019). This is a threshold for the rate of growth of the real wage rate such that the following holds: if the growth rate of the real wage rate is higher than the threshold, the rate of profit will fall (and Marx's claim about the falling rate of profit will hold); if the growth rate of the real wage rate is lower than the threshold, the rate of profit will rise (and Okishio's claim about the rising rate of profit will be true).

To derive the value of the Marx-Okishio threshold, let us assume that the real wage rate is αb after the new technique of production is adopted, where α is some real number, that is, the real wage rate increases by $100(\alpha - 1)$ per cent. In this new situation, the rate of profit is

$$r''' = \frac{\text{profit}}{\text{total outlay}} = \frac{1 - a' - \alpha b n'}{a' + \alpha b n'} = \frac{1}{a' + \alpha b n'} - 1$$

The Marx-Okishio threshold is the value of α that ensures that $r''' = r$, that is, the rate of profit remains unchanged even after the new technique of

production has been adopted. Let us denote the Marx-Okishio threshold as α^* , and note that it must satisfy the following condition,

$$\frac{1}{a' + \alpha^* bn'} - 1 = \frac{1}{a + bn} - 1$$

so that the Marx-Okishio threshold, α^* , is given by

$$\alpha^* = \frac{n}{n'} + \frac{a - a'}{bn'} = \frac{n}{n'} + \left(\frac{a}{a'} - 1 \right) \frac{a'}{bn'} \quad (6.7)$$

How would a researcher calculate the Marx-Okishio threshold? Define n and n' as the reciprocal of output per person before and after technical change, respectively; define a and a' as the ratio of capital stock and net value added; finally, note that bn' is the share of wages in real value added. Once we collect data on these variables, we can use (6.7) to calculate the Marx-Okishio threshold.

Proposition 3. *If the real wage rate increases by more than $100(\alpha^* - 1)$ per cent after the new technique of production is adopted, then the rate of profit will fall; if the real wage rate increases by less than $100(\alpha^* - 1)$ per cent, then the rate of profit will rise.*

The proof follows immediately from the preceding discussion. It is important to note that this proposition shows that both Marx and Okishio are right. What happens to the rate of profit after technical change in a capitalist economy depends on the movement of the real wage rate. If the real wage rate is sluggish, then the rate of profit will rise after technical change; if the real wage rate rises relatively rapidly, then the rate of profit will fall. Without more knowledge about the trajectory of the real wage rate, it is not possible to unambiguously predict the effect of technical change on the average rate of profit.

A useful way to summarize the whole argument presented in this section is to break down the process of technical change into four steps and see what happens to the profit rate at each step.

- Step 1: This is the initial situation. All capitalists use the old technique of production. The average profit rate is given by r .
- Step 2: A new technique becomes available to an innovator capitalist, and she evaluates whether it is profitable to adopt it. When carrying

out this evaluation, she uses the existing prices and real wage rate. If the viability condition is satisfied, she adopts the new technique and her profit rate is r' . Obviously, the rate of profit she earns is higher than before, that is, $r' > r$. It is important to keep in mind that r' is the individual rate of profit (earned only by the innovator capitalist.)

- Step 3: Other capitalists adopt the new technique because of diffusion of the new technique across the economy. The average rate of profit, r'' , is now equal to r' because all capitalists use the same technique of production as the innovator capitalist. Hence, the average rate of profit is higher than in the initial situation, that is, when the old technique was being used, that is, $r'' > r$. (Note that $r'' = r'$ in the one-commodity model I am using. If I used an n -commodity model, then r'' would not in general be equal to r' .)
- Step 4: There are changes in the labour market (or changes in the balance of class forces). This leads to a change in the real wage rate. If $\alpha > \alpha^*$, that is, if the real wage rate rises by more than the Marx-Okishio threshold, then the average profit rate falls, that is, $r''' < r$, where r''' is the new economy-wide, average rate of profit.

The last step, whereby the rate of profit falls, highlights an important problem faced by capitalist economies. This relates to the divergence between actions that are individually rational from those that are collectively rational. In this case, it is individually rational for a capitalist innovator to search for and adopt new techniques of production that reduce the cost of production. But under certain conditions, adopting the new technique reduces the average rate of profit and goes against the collective interests of the capitalist class.

The problem is that capitalism suffers from a *social coordination* problem (Foley et al., 2019). Being a competitive system, there is no way that capitalists can coordinate their actions. Once the innovator capitalist adopts the new technique, her rate of profit rises. Competitive pressures then lead to a diffusion of the new technique through the rest of the economy. What the individual capitalist, and the capitalist class as a whole, does not have control over is the effect of technical change on the average real wage rate. In the case when the institutional set-up allows for workers to bargain for and get a relatively higher real wage rate, the average rate of profit falls. Individually rational actions become collectively irrational!

Let us use data from the US CB sector to see the empirical relevance of the Marx-Okishio threshold. In Figure 6.2, I present a scatter plot of two variables measured annually for the period between 1979 and 2019. On the horizontal axis, I measure the difference between the ratio of the average weekly real wage rate in successive years and the Marx-Okishio threshold for each year (calculated using the expression in (6.7)); on the vertical axis, I measure the change in the annual rate of profit (measured as the ratio of net operating surplus and the replacement cost value of net fixed assets). From Figure 6.2, we see a strong negative relationship between the variables: when the actual real wage rate is higher than the Marx-Okishio threshold, the rate of profit declines, which is exactly what we expect on the basis of the discussion in this section. The average relationship is captured by the dashed line (the bivariate regression line) which has a statistically significant slope of -1.88. This means that, on average, if the ratio of actual real wage rate is higher than the Marx-Okishio threshold by unity, then the rate of profit declines by 1.88 percentage points.

6.4.1 *Marx-Okishio Threshold for Different Types of Technical Change*

To see how the Marx-Okishio threshold changes according to the type of technical change, let us go back to (6.7). Using the expression for the Marx-Okishio threshold, we see that,

$$\alpha^* - 1 = \left(\frac{n}{n'} - 1 \right) + \left(\frac{a}{a'} - 1 \right) \frac{a'}{bn'} \quad (6.8)$$

is composed of three parts: the rate of change of labour productivity,

$$\frac{(1/n' - 1/n)}{(1/n)} = \frac{n}{n'} - 1$$

the rate of change of capital productivity,

$$\frac{(1/a' - 1/a)}{(1/a)} = \frac{a - a'}{a'} = \left(\frac{a}{a'} - 1 \right)$$

and the composition of capital of the new technique of production at the original real wage rate,

$$\frac{a'}{bn'}$$

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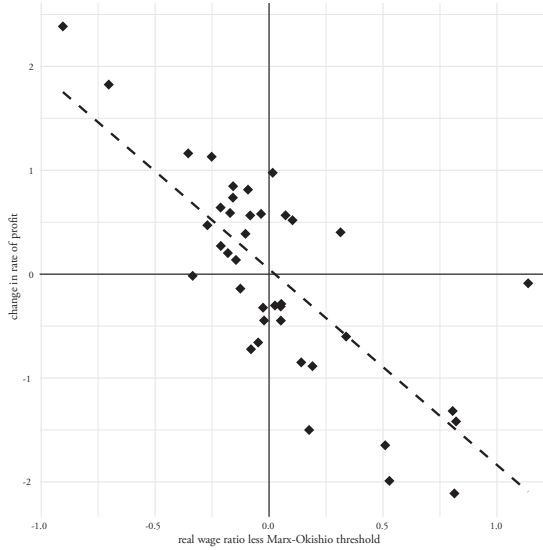


Figure 6.2 Scatter plot for the US CB sector between 1979 and 2019 of (a) the difference between ratio of real wage rate in consecutive years and the Marx-Okishio threshold for that year and (b) change in the rate of profit

Source: Author's calculations using data from NIPA Table 1.14, US Bureau of Economic Analysis, and FRED, Federal Reserve Bank of St. Louis.

For CS-LS technical change, since $0 < a' < a$ and $0 < n' < n$, all the terms on the right-hand side of (6.8) are positive, that is, $\alpha^* > 1$.¹⁴ This means that there is a whole range of positive growth rates of real wage rate, including, of course, zero real wage growth, that is, compatible with a rising rate of profit: this range is given by $0 \leq \alpha < \alpha^*$, where $\alpha - 1$ is the actual growth rate of the real wage rate. Only if the actual growth rate of the real wage is larger than $\alpha^* - 1$ will the rate of profit fall.

On the other hand, CU-LS technical change highlights a trade-off that capitalists face. Here, technical change reduces the labour input, but only at the cost of increasing the capital input. This is a case that was discussed by Marx and has been highlighted in the work of Duncan Foley and Tom Michl, who call it Marx-biased technical change (Foley and Michl, 1999). It opens up the possibility that technical change, the key strength of capitalism, might

¹⁴We rule out $a' = 0$ to avoid division by zero.

also reduce profitability, the driving force of capitalism. This is, therefore, a peculiar way in which capitalism undermines itself.

To see this, let us note that the last expression on the right-hand side of (6.8) is negative for CU-LS technical change (this is because $a' > a$). If we also have

$$\left(\frac{n}{n'} - 1\right) < \left(1 - \frac{a}{a'}\right) \frac{a'}{bn'}$$

then this shows that $\alpha^* - 1 < 0$. This condition tells us that if the growth rate of labour productivity, $(n/n') - 1$, is less than negative of the change in capital productivity, $1 - (a/a')$, multiplied by the OCC of the new technique (evaluated at old prices), $(a'/bn') = (pa'/wn')$, then $\alpha^* - 1 < 0$. In such a case, we have a situation where *any* increase in the actual real wage rate, including a zero growth, will lead to a fall in the rate of profit. The relevant range is captured by the condition $\alpha^* < 1 \leq \alpha$, where $\alpha - 1$ is the actual growth rate of the real wage rate. In fact, it is striking to note that there is a whole range of negative values of the growth rate of the real wage rate that is also compatible with a fall in the rate of profit. This range is given by $\alpha^* < \alpha < 1$, where $\alpha - 1$ is the actual growth rate of the real wage rate. Only if the real wage rate is *pushed down* far enough, that is, $\alpha < \alpha^*$, will the rate of profit rise. Of course, in this case, the capital-labour conflict over the distribution of income is bound to be intense.

At the end of the previous section, we had computed the Marx-Okishio threshold for the US CB sector between 1979 and 2019. When we compute the average value of the Marx-Okishio threshold between 2000 and 2019, we get 0.915. Thus, here we have a case where the Marx-Okishio threshold is lower than unity. In such a situation, even a zero growth of the real wage rate would have led to a decline in the rate of profit.

6.5 Constant Rate of Exploitation

We have seen earlier that if the wage rate grows at a rate that is larger than the Marx-Okishio threshold, then the rate of profit will fall after technical change. Here we wish to investigate a special case, one that Marx used in his analysis, where the rate of exploitation remains constant before and after technical change. This is a special case of the analysis with a rising real wage rate because, in a context of rising labour productivity, the rate of

exploitation can remain constant only if the real wage rate rises. It is under this assumption that Marx had worked out his LTFRP (Marx, 1993b, Chapter 13). The next proposition shows that Marx's claim is correct even when we consider capitalistically viable technical change.

Proposition 4. *Consider capitalistically viable technical change that is of the CU-LS type. If the rate of exploitation does not change, then the rate of profit will fall.*

To see the proof, recall that the rate of exploitation, denoted by e , is the ratio of surplus value and variable capital. Let us denote by λ the value of one unit of corn when the old technique of production, (n, a) , is in use. As before, let us assume that the real wage rate is b . Then, the variable capital per hour of labour is $nb\lambda$ and the surplus value is $(\lambda - (nb + a)\lambda)$. Hence, the rate of exploitation is given by

$$e = \frac{1 - (nb + a)}{nb} \quad (6.9)$$

Suppose the new technique of production is given by (n', a') and the value of a unit of corn using this new technique of production is given by λ' . If the real wage rate changes to αb , an argument similar to the one used above shows that the rate of exploitation, after technical change, is given by

$$e' = \frac{1 - (n'\alpha b + a')}{n'\alpha b}$$

For the rate of exploitation to be constant, that is, $e' = e$, the value of α must be given by:

$$\bar{\alpha} = \frac{(1 - a')n}{(1 - a)n'} \quad (6.10)$$

To see what happens to the rate of profit, in this case, we need only compare $\bar{\alpha}$ with α^* , the Marx-Okishio threshold given in (6.7). A little algebraic manipulation shows that

$$\bar{\alpha} - \alpha^* = \frac{(bn + a - 1)(a - a')}{nb(1 - a)} \quad (6.11)$$

Since $a < 1$, the denominator of the expression on the right-hand side is positive. Since technical change is of the CU-LS type, we have $a' > a$ and

$n' < n$. Hence, $a - a' < 0$. Moreover, capitalist producers will only adopt techniques of production that are productive. This applies to the old technique of production, as much as to the new. Hence, $bn + a < 1$. Hence, both terms in the numerator of (6.11) are negative. Since the denominator is positive, this shows that the fraction is positive, so that $\bar{\alpha} > \alpha^*$. Using Proposition 3, we see that the rate of profit will fall after technical change.

What is the intuitive meaning of this result? Proposition 4 tells us that when a capitalist economy undergoes CU-LS technical change and socio-economic forces manage to keep the rate of exploitation (or the ratio of profit and wage income) constant, then the rate of profit will fall. How can this come about? Let us imagine a new cost-reducing technique of production being adopted by an innovator capitalist. As the new technique diffuses in the capitalist economy, other producers adopt it too. Over time, the new technique becomes universally used. Parallel to this process of technical change, the economy also witnesses changes in the labour market. In the context of the rise in labour productivity coming from the adoption of the new technique of production, workers are able to bid up the real wage rate. In fact, they manage to push up the real wage to the extent that is necessary to keep the rate of exploitation (or the profit-wage ratio) constant, that is, to keep the balance of class forces unchanged.

If workers manage to do so, the new technique of production, which has already been universally adopted, is no longer cost-reducing when compared to the original technique of production. This last point can be seen by comparing the cost of production with the new technique of production at the new real wage rate, with the cost of production with the original technique of production at the old real wage rate:

$$(n'\alpha b + a') - (nb + a)$$

Plugging the value of α from (6.10), we get

$$\begin{aligned} n'\alpha b + a' - (nb + a) &= \frac{bn(1 - a')}{(1 - a)} + a' - nb - a \\ &= \frac{(a' - a)(1 - a - nb)}{1 - a} > 0 \end{aligned}$$

because technical change is CU-LS, so that $a' - a > 0$, and the original technique of production is productive, so that $1 - a - nb > 0$. That is why the rate of profit falls.

6.6 Conclusion

In this chapter, we have studied the relationship between capitalism and technical change. We have highlighted two important results. First, technical change in a capitalist context is crucially impacted by profitability considerations because only those new techniques of production are adopted that are capitalistically viable. This creates the possibility of divergence between societal needs and capitalist profitability. Many techniques of production that are desirable from a societal point of view will not be adopted because they do not increase the rate of profit. Similarly, many new techniques of production might be adopted that are inimical to broader societal interests.

Second, we have presented an approach to the vexed issue of the impact of technical change on the rate of profit. We have argued that there is no necessary contradiction between the claims advanced by Marx and Okishio. Both can be accommodated in a more general framework where we allow the real wage rate to be impacted by class struggle. In that case, which seems quite realistic, whether or not the rate of profit falls after technical change in a capitalist economy will depend crucially on whether workers are able to force a rise in the real wage rate parallel to technical change.

One special case of this general discussion regarding the effect of technical change on the rate of profit was discussed by Marx in Volume III of *Capital*. This is the case where the rate of exploitation (also the wage–profit ratio) remains constant over the process of technical change. This is not implausible. In many advanced capitalist countries, institutions of collective bargaining might allow workers to keep the wage share of national income, that is, the wage–profit ratio, constant. In this institutional context, if technical change is of the CU-LS type, which Foley and Michl (1999) have called Marx-biased technical change, then the average rate of profit will fall.

There is an interesting technical issue that is worth highlighting. In the one-commodity model that we have worked with in this chapter, the rate of exploitation is also equal to the ratio of profits and wages. To see this, note that profit income per unit of output is $p - p(nb + a)$ and nominal wages per unit of output is pnb . Hence, the ratio of profits and wages is equal to e , the rate of exploitation, as defined in (6.9). The identity of the rate of exploitation, which is defined in terms of value, and the profit–wage ratio, which is defined in terms of prices, is no longer true in models with more than one commodity. In fact,

in general linear models, where the nominal wage rate is the numeraire, the scale of production in each sector enters into the definition of the profit–wage ratio (Roemer, 1981, Chapter 6). On the other hand, the rate of exploitation can be defined without referring to the scale of production. Hence, in general, the two will not be equal. There is one special case, the New Interpretation of Marx’s labour theory of value, which I discuss in detail in the next chapter, where the equality obtains even in a general n -commodity model.

Further Readings

- An insightful discussion of various aspects of technical change can be found in Laibman (1992, Chapters 5–8). In these chapters, the effect of technical change on the rate of profit when the rate of exploitation remains unchanged is worked out in a two-commodity model.
- A rigorous treatment of the issues discussed in this section within the context of a n -commodity model can be found in Roemer (1981, Chapters 4–6).
- An analysis of the effect of technical change on the rate of profit when the profit–wage ratio remains constant is worked out in a two-commodity model in Roemer (1977, 1981). An important finding is that only sectoral profit–wage ratios, but not the aggregate profit–wage ratio, can be defined without reference to the scale of production. Moreover, sectoral profit–wage ratios can remain constant only when the real wage varies across sectors, that is, we need to assume non-competitive labour markets. The main result in Roemer (1977, 1981) is that the rate of profit falls (or remains unchanged) if there is cost-reducing CU-LS technical change in the capital goods (consumer goods) sector. This result has been generalized to the case of a n -commodity model in Chen (2019), which shows that when there is cost-reducing CU-LS technical change in any sector with sectoral profit–wage ratios remaining constant, the equilibrium profit rate falls.